

IP PLAN:

IP ADDRESSING TABLE

R1 (ROAS + IP Helper)

G0/0.10	VLAN 10	192.168.10.1	255.255.255.0	Gateway + Helper
G0/0.20	VLAN 20	192.168.20.1	255.255.255.0	Gateway + Helper
G0/0.30	VLAN 30	192.168.30.1	255.255.255.0	Gateway + Helper
G0/0.99	VLAN 99	192.168.99.1	255.255.255.0	Native + Gateway + Helper
G0/1	10.0.0.1	255.255.255.252	to R2	

R2 (DHCP)

G0/0	10.0.0.3	255.255.255.252	to R1
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S1 (Root Bridge)

VLAN 99	192.168.99.2	255.255.255.0	SVI (Mgmt)
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Default Gateway: 192.168.99.1

S2

VLAN 99	192.168.99.3	255.255.255.0	SVI (Mgmt)
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Default Gateway: 192.168.99.1

END DEVICES

PC0	VLAN 10	192.168.10.10	255.255.255.0	GW 192.168.10.1
PC1	VLAN 10	192.168.10.11	255.255.255.0	GW 192.168.10.1
PC2	VLAN 20	192.168.20.10	255.255.255.0	GW 192.168.20.1
PC3	VLAN 20	192.168.20.11	255.255.255.0	GW 192.168.20.1
Server0	VLAN 30	192.168.30.10	255.255.255.0	GW 192.168.30.1 (Static)

VLAN SUMMARY

VLAN 10 → 192.168.10.0/24
VLAN 20 → 192.168.20.0/24
VLAN 30 → 192.168.30.0/24
VLAN 99 → 192.168.99.0/24 (Native) (Management)
VLAN 666 → BlackHole (no IP)

INTER-ROUTER LINK

10.0.0.0/30
R1 = 10.0.0.1
R2 = 10.0.0.2

Device Identity & Basic Security

S1:

```
S1#show running-config | include hostname|enable secret|username|ip domain-name|ip ssh|banner
hostname S1
enable secret 5 $1$mERr$tUYWAIdTbI.xoc21/o.Eg1
ip ssh version 2
ip domain-name OYL.com
username admin secret 5 $1$mERr$tUYWAIdTbI.xoc21/o.Eg1
banner motd ^C
S1#show users
  Line      User      Host(s)      Idle      Location
*  0 con 0    admin      idle         00:00:00

  Interface  User      Mode      Idle      Peer Address
S1#
```

S2:

```
S2#
S2#show running-config | include hostname|enable secret|username|ip domain-name|ip ssh|banner
hostname S2
enable secret 5 $1$mERr$tUYWAIdTbI.xoc21/o.Eg1
ip ssh version 2
ip domain-name OYL.com
username admin secret 5 $1$mERr$tUYWAIdTbI.xoc21/o.Eg1
banner motd ^C
S2#show users
  Line      User      Host(s)      Idle      Location
*  0 con 0    admin      idle         00:00:00

  Interface  User      Mode      Idle      Peer Address
S2#
```

R1:

```
R1#
R1#show running-config | include hostname|enable secret|username|ip domain-name|ip ssh|banner
hostname R1
enable secret 5 $1$mERr$tUYWAIdTbI.xoc21/o.Eg1
username admin secret 5 $1$mERr$tUYWAIdTbI.xoc21/o.Eg1
ip domain-name OYL.com
banner motd ^C
R1#show users
  Line      User      Host(s)      Idle      Location
*  0 con 0    admin      idle         00:00:00

  Interface  User      Mode      Idle      Peer Address
R1#
```

R2:

R2#

```
R2#show running-config | include hostname|enable secret|username|ip domain-name|ip ssh|banner
hostname R2
enable secret 5 $1$mERr$tUYWAIdTbI.xoc21/o.Eg1
username admin secret 5 $1$mERr$tUYWAIdTbI.xoc21/o.Eg1
ip ssh version 2
ip domain-name OYL.com
banner motd ^C
```

R2#show users

Line	User	Host(s)	Idle	Location
* 0 con 0	admin	idle	00:00:00	

Interface	User	Mode	Idle	Peer Address
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R2#

VLAN	VLAN Name	Status	Ports
	1 default	active	
	10 vlan-10	active	Fa0/10, Fa0/11
	20 vlan-20	active	
	30 vlan-30	active	
S1:	99 vlan-99-native	active	
	666 vlan-666-blackhole	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/2
S2:	1002 fddi-default	active	
	1003 token-ring-default	active	
	1004 fddinet-default	active	
	1005 trnet-default	active	

S1#show vlan id 10

VLAN Name	Status	Ports
10 vlan-10	active	Fa0/10, Fa0/11

VLAN Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
10	enet	100010	1500	-	-	-	-	0	0

S1#show vlan id 20

VLAN Name	Status	Ports
20 vlan-20	active	

VLAN Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
20	enet	100020	1500	-	-	-	-	0	0

S1#show vlan id 30

VLAN Name	Status	Ports
30 vlan-30	active	

VLAN Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
30	enet	100030	1500	-	-	-	-	0	0

S1#show vlan id 99

VLAN Name	Status	Ports
99 vlan-99-native	active	

VLAN Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
99	enet	100099	1500	-	-	-	-	0	0

S1#show vlan id 666

VLAN Name	Status	Ports
666 vlan-666-blackhole	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/2

VLAN Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
666	enet	100666	1500	-	-	-	-	0	0

VLAN	Name	Status	Ports
1	default	active	
10	vlan-10	active	
20	vlan-20	active	Fa0/20, Fa0/21
30	vlan-30	active	Fa0/24
99	vlan-99-native	active	
666	vlan-666-blackhole	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/22, Fa0/23, Gig0/1 Gig0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

S2#show vlan id 10

VLAN	Name	Status	Ports
10	vlan-10	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
10	enet	100010	1500	-	-	-	-	-	0	0

S2#show vlan id 20

VLAN	Name	Status	Ports
20	vlan-20	active	Fa0/20, Fa0/21

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
20	enet	100020	1500	-	-	-	-	-	0	0

S2#show vlan id 30

VLAN	Name	Status	Ports
30	vlan-30	active	Fa0/24

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
30	enet	100030	1500	-	-	-	-	-	0	0

S2#show vlan id 99

VLAN	Name	Status	Ports
99	vlan-99-native	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
99	enet	100099	1500	-	-	-	-	-	0	0

S2#show vlan id 666

VLAN	Name	Status	Ports
666	vlan-666-blackhole	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/22, Fa0/23, Gig0/1 Gig0/2

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
666	enet	100666	1500	-	-	-	-	-	0	0

EtherChannel Status

S1:

```
S1#show etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SU)          LACP       Fa0/1(P) Fa0/2(P)
S1#
```

S2:

```
S2#show etherchannel summary
Flags:  D - down          P - in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SU)          LACP       Fa0/1(P) Fa0/2(P)
S2#
```

Trunks

S1:

```
S1#show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Po1       on        802.1q         trunking      1
Gig0/1    on        802.1q         trunking      1

Port      Vlans allowed on trunk
Po1       10,20,30,99
Gig0/1    10,20,30,99

Port      Vlans allowed and active in management domain
Po1       10,20,30,99
Gig0/1    10,20,30,99

Port      Vlans in spanning tree forwarding state and not pruned
Po1       10,20,30,99
Gig0/1    10,20,30,99
```

S2:

```

S2#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Po1       on        802.1q         trunking    1

Port      Vlans allowed on trunk
Po1       10,20,30,99

Port      Vlans allowed and active in management domain
Po1       10,20,30,99

Port      Vlans in spanning tree forwarding state and not pruned
Po1       10,20,30,99

S2#

```

Port-Security

S1:

```

S1#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
          (Count)      (Count)      (Count)
-----
Fa0/10    3          1          0          Shutdown
Fa0/11    3          1          0          Shutdown
-----

S1#
S1#
S1#
S1#
S1#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
          (Count)      (Count)      (Count)
-----
Fa0/10    3          1          0          Shutdown
Fa0/11    3          1          0          Shutdown
-----

S1#

```

S2:

```

S2#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Po1       on        802.1q         trunking    1

Port      Vlans allowed on trunk
Po1       10,20,30,99

Port      Vlans allowed and active in management domain
Po1       10,20,30,99

Port      Vlans in spanning tree forwarding state and not pruned
Po1       10,20,30,99

S2#
S2#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
          (Count)      (Count)      (Count)
-----
Fa0/20    3          1          0          Shutdown
Fa0/21    3          1          0          Shutdown
Fa0/24    3          1          0          Shutdown
-----

S2#

```

PortFast + BPDU Guard

S1:

```
S1#show spanning-tree summary
Switch is in pvst mode
Root bridge for: default vlan-10 vlan-20 vlan-30 vlan-99-native
Extended system ID      is enabled
Portfast Default        is disabled
PortFast BPDU Guard Default is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default       is disabled
EtherChannel misconfig guard is disabled
UplinkFast              is disabled
BackboneFast            is disabled
Configured Pathcost method used is short
```

Name	Blocking	Listening	Learning	Forwarding	STP Active
VLAN0001	0	0	0	4	4
VLAN0010	0	0	0	6	6
VLAN0020	0	0	0	4	4
VLAN0030	0	0	0	4	4
VLAN0099	0	0	0	4	4
6 vlans	0	0	0	22	22

```
S1#
```

S2:

```
S2#show spanning-tree summary
Switch is in pvst mode
Root bridge for:
Extended system ID      is enabled
Portfast Default        is disabled
PortFast BPDU Guard Default is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default       is disabled
EtherChannel misconfig guard is disabled
UplinkFast              is disabled
BackboneFast            is disabled
Configured Pathcost method used is short
```

Name	Blocking	Listening	Learning	Forwarding	STP Active
VLAN0001	0	0	0	3	3
VLAN0010	0	0	0	3	3
VLAN0020	0	0	0	5	5
VLAN0030	0	0	0	4	4
VLAN0099	0	0	0	3	3
6 vlans	0	0	0	18	18

```
S2#
```

Management

S1:

```
S1#
S1#show ip interface brief | include Vlan|Port-channel
Port-channel1      unassigned      YES manual up      up
Vlan1               unassigned      YES manual administratively down down
Vlan10             unassigned      YES manual up      up
Vlan20             unassigned      YES manual up      up
Vlan30             unassigned      YES manual up      up
Vlan99             192.168.99.99   YES manual up      up
Vlan666            unassigned      YES manual up      down
S1#show ip default-gateway
```

Router Subinterfaces

R1:

```
R1#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0 10.0.0.2        YES manual up        up
GigabitEthernet0/0.99 unassigned      YES unset up        up
GigabitEthernet0/1 unassigned      YES unset up        up
GigabitEthernet0/1.10 192.168.10.1    YES manual up        up
GigabitEthernet0/1.20 192.168.20.1    YES manual up        up
GigabitEthernet0/1.30 192.168.30.1    YES manual up        up
GigabitEthernet0/1.99 unassigned      YES unset up        up
GigabitEthernet0/2 unassigned      YES unset administratively down down
Vlan1              unassigned      YES unset administratively down down
R1#
```

Routing:

R1:

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/30 is directly connected, GigabitEthernet0/0
L    10.0.0.2/32 is directly connected, GigabitEthernet0/0
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.10.0/24 is directly connected, GigabitEthernet0/1.10
L    192.168.10.1/32 is directly connected, GigabitEthernet0/1.10
192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.20.0/24 is directly connected, GigabitEthernet0/1.20
L    192.168.20.1/32 is directly connected, GigabitEthernet0/1.20
192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.30.0/24 is directly connected, GigabitEthernet0/1.30
L    192.168.30.1/32 is directly connected, GigabitEthernet0/1.30
R1#
```

R2:

```

R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
S       192.168.10.0/24 [1/0] via 10.0.0.2
S       192.168.20.0/24 [1/0] via 10.0.0.2
S       192.168.30.0/24 [1/0] via 10.0.0.2
R2# |

```

DHCP:

R2:

```

R2#show ip dhcp pool

Pool VLAN-10 :
Utilization mark (high/low)      : 100 / 0
Subnet size (first/next)          : 0 / 0
Total addresses                   : 254
Leased addresses                  : 2
Excluded addresses                : 3
Pending event                     : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
192.168.10.1      192.168.10.1 - 192.168.10.254    2 / 3 / 254

Pool VLAN-20 :
Utilization mark (high/low)      : 100 / 0
Subnet size (first/next)          : 0 / 0
Total addresses                   : 254
Leased addresses                  : 2
Excluded addresses                : 3
Pending event                     : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
192.168.20.1      192.168.20.1 - 192.168.20.254    2 / 3 / 254

Pool VLAN-30 :
Utilization mark (high/low)      : 100 / 0
Subnet size (first/next)          : 0 / 0
Total addresses                   : 254
Leased addresses                  : 0
Excluded addresses                : 3
Pending event                     : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
192.168.30.1      192.168.30.1 - 192.168.30.254    0 / 3 / 254
R2#show ip dhcp binding
IP address      Client-ID/      Lease expiration      Type
                Hardware address
192.168.10.2    00E0.F9CA.7485      --                     Automatic
192.168.10.3    0001.432B.4C8C      --                     Automatic
192.168.20.2    0010.110D.BDDA      --                     Automatic
192.168.20.3    0001.9651.913E      --                     Automatic
R2#

```

IP-Helper:

R1:

```
R1#show running-config | include helper-address  
ip helper-address 10.0.0.1  
ip helper-address 10.0.0.1  
ip helper-address 10.0.0.1  
ip helper-address 10.0.0.1  
R1#
```
